Internal Capital Frictions in Intermediaries and Exchange Rate Dynamics

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Intermediary Asset Pricing: A Primer

Starting point in Macroeconomics: the **Consumption Capital Asset Pricing Model**

$$\text{(EE) } U'(C_t) = \mathbb{E}_t\left[\beta U'(C_{t+1})R_{t+1}\right] \implies \mathbb{E}_t\left[M_{t+1}R_{t+1}\right] = 1,\ M_{t+1} = \beta\frac{U'(C_{t+1})}{U'(C_t)}\ \text{(SDF)}$$

Problem: real consumption is too stable, leading to the Excess Volatility Puzzle (Shiller, 1981) and Equity Premium Puzzle (Mehra & Prescott, 1985).

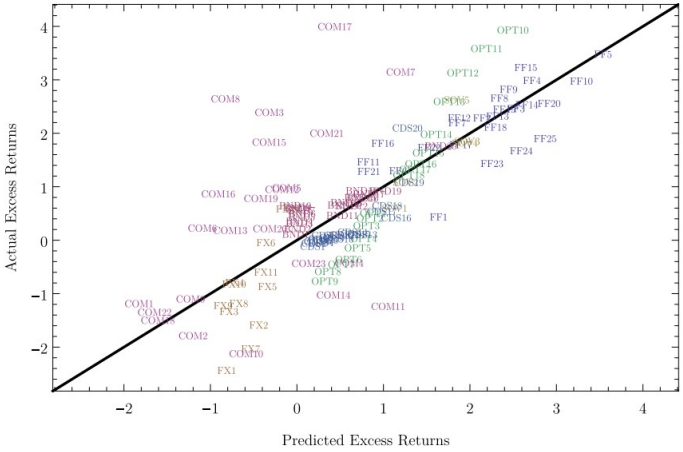
(One) Solution: **Intermediary Asset Pricing Theory**.

Segmented markets: households invest through an intermediary. But financial intermediation is imperfect: moral hazard, adverse selection, asymmetric information and monitoring costs...

Theory: it is about the equity/net worth of intermediaries, need bankers to have skin in the game (Incentive Compatibility Constraint).

Intermediary Asset Pricing: A Primer (2)

Prediction: it is the SDF of intermediaries, not households, that prices financial assets.
Empirics: intermediary leverage (He, Kelly and Manela, 2017; Adrian, Etula and Muir, 2014).



This project in a nutshell

- **Motivation:** Reconcile the Intermediary Asset Pricing theory (equity) with the empirics (leverage of the financial sector), consistently with the pecking order (internal funds before external ones).
- **Research question:** Can internal funding frictions explain CIP violations, thus better approximating the intermediaries' SDF?
- **Key mechanism:** A liquidity requirement elsewhere for the intermediary limits arbitrage in currency pairs in which the intermediary is dominant, driving CIP wedges.
- **Data:** (UK-)EMIR data of all the derivative trades in the UK, including FX forwards.
- **Empirical strategy:** Explain CIP deviations with shocks to the profit and loss statement (P&L) of big banks/broker-dealers that intermediate the market for FX forwards.

This project in a nutshell

- **Motivation:** Reconcile the Intermediary Asset Pricing theory with the empirics, consistently with the pecking order.
- **Research question:** Can variation margins and internal funding frictions explain CIP violations, thus better approximating the stochastic discount factor of financial intermediaries?
- **Theoretical model:** Variation margins create liquidity risk that banks price ex-ante.
- **Data:** (UK-)EMIR data of all the derivative trades in the UK, including FX forwards.
- **Empirical strategy:** Explain CIP deviations with shocks to the profit and loss statement (P&L) of big banks/broker-dealers that intermediate the market for FX forwards.

Our contributions to the literature

We study internal liquidity and CIP violations, proxy for intermediaries' SDF

- Theory: Shocks to intermediaries can be important, especially in the context of FX determination
- Empirically, many limits to arbitrage can explain CIP violations
- ➤ Because of limits to arbitrage, prices can move away from fundamentals

We empirically document the market microstructure for FX forwards

- ➤ The market is OTC and still intermediated by large dealers

We use individual P&L changes as shocks to explain aggregate prices

- ➤ Idiosyncratic shocks can have effects on macroeconomic variables

Our contributions to the literature

We study internal liquidity and CIP violations, proxy for intermediaries' SDF

- Theory:
  - Intermediary Asset Pricing: Equity (Bernanke and Gertler, 1989; He and Krishnamurthy, 2012, 2013, 2018), Debt (Kiyotaki and Moore, 1997; Brunnermeier and Pedersen, 2009) Leverage (Holmstrom and Tirole, 1997; Aiyagari and Gertler, 1999; Gertler and Karadi, 2011; Brunnermeier and Sannikov, 2014).
  - The effect can be important for arbitrage as well: Shleifer and Vishny (1997), Gromb and Vayanos (2002, 2010, 2018).
  - Exchange rate determination: Gabaix and Maggiori (2015), Itskhoki and Mukhin (2021, 2024).
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  Empirics:
  - Leverage of broker-dealer sector matters: Adrian, Etula, and Muir (2014), He, Kelly, and Manela (2017).
  - [Pre-GFC] Small deviations because of **credit risk** and transaction costs, capital controls.
  - **Balance-sheet constraints** on regulated arbitrageurs: Du et al. (2018), Cenedese et al. (2021); also impact unregulated ones: Boyarchenko et al. (2018)
  - **Funding segmentation**: Sushko et al. (2016), Rime et al. (2022), Anderson et al. (2021).
  - **Margins**: Gârleanu and Pedersen (2011), Augustin et al. (2024).
  - Growing evidence that **internal funding** matters: Duffie (2010), Siriwardane (2019).

➤ Because of limits to arbitrage, prices can move away from fundamentals

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- Search and matching for ad-hoc demand requires market dealers: Weill (2020). Alternatively, focus on the role of information: Collin-Dufresne et al. (2019).
- Financial networks and price formation: Hau et al. (2021), Eisfeldt et al. (2022).
- Recent changes: move away from phone orders (!), RFQ / MTF / SI, algorithmic trading.
- ➤ The market is OTC and still intermediated by large dealers

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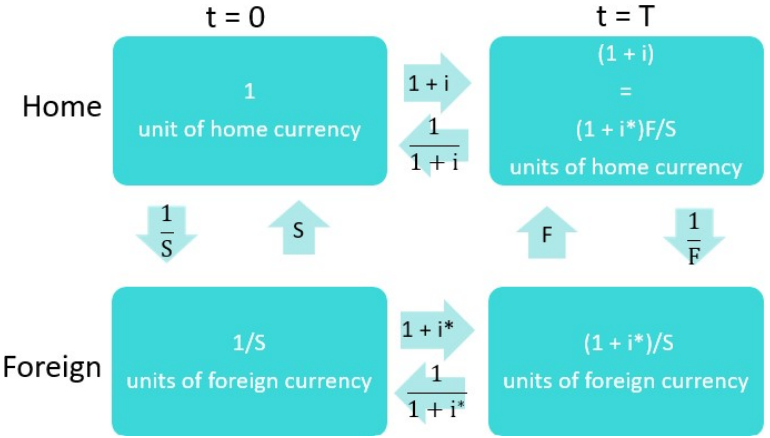
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- The market is OTC and still intermediated by large dealers

We use individual P&L changes as shocks to explain aggregate prices

- Internal capital constraints for trading desks with mark-to-market losses acts as segmentation: Mitchell, Pedersen, and Pulvino (2007)
- Arbitrage is driven by a limited number of market participants: Siriwardane et al. (2023).
- Granular Instrumental Variables: Gabaix and Koijen (2020).
- Idiosyncratic shocks can have effects on macroeconomic variables

The Covered Interest Rate Parity (CIP) - Theory



Pricing an FX swap:

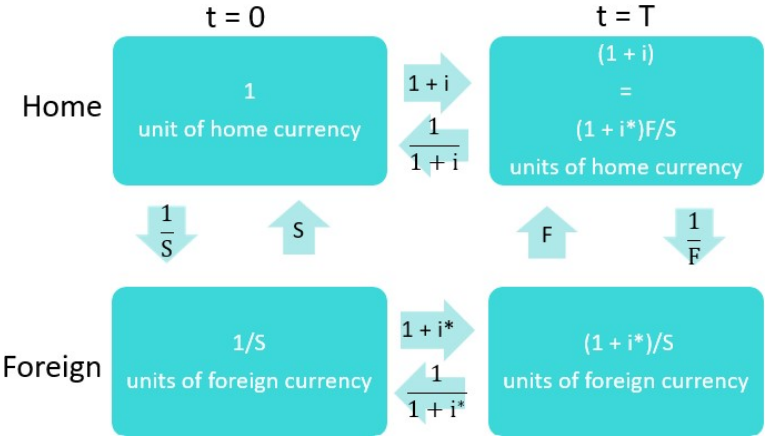
$$1 + i_{0,T} = \left(1 + i^*_{0,T}\right) \frac{F_{0,T}}{S_0}$$

$$\Leftrightarrow$$

$$F_{0,T} = S_0 \frac{1 + i_{0,T}}{1 + i^*_{0,T}}$$

where i is the interest rate in Home, i^* is the interest rate in Foreign, F is the forward rate Foreign/Home, S is the spot rate Foreign/Home.

The Covered Interest Rate Parity (CIP) - Theory



Pricing an FX swap:

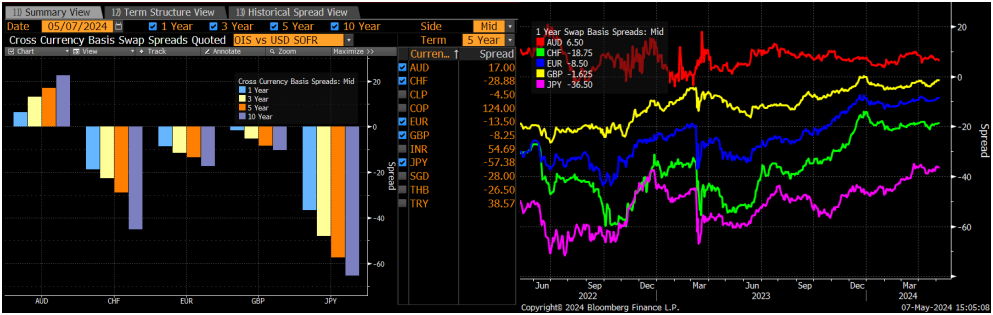
$$1 + i_{0,T} = \left(1 + i^*_{0,T}\right) \frac{F_{0,T}}{S_0}$$

$$\Leftrightarrow$$

$$F_{0,T} = S_0 \frac{1 + i_{0,T} + b_T}{1 + i^*_{0,T}}$$

where i is the interest rate in Home, i^* is the interest rate in Foreign, F is the forward rate Foreign/Home, S is the spot rate Foreign/Home, b_T is the cross-currency basis.

The Covered Interest Rate Parity (CIP) - Empirics



(a) Bloomberg XCCY menu (b) Cross currency basis swap

The key identification hypothesis relies on the no-arbitrage condition: the basis is free money for market participants able to enter this market, and as such any fundamental driver can only affect the basis through the intermediaries.

Funds Transfer Pricing (FTP)

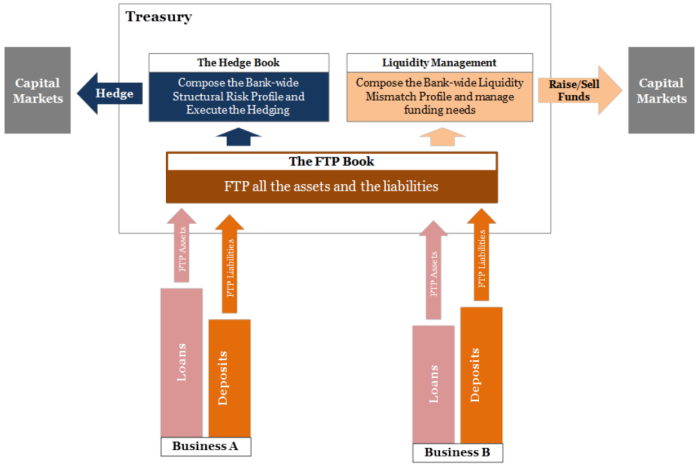
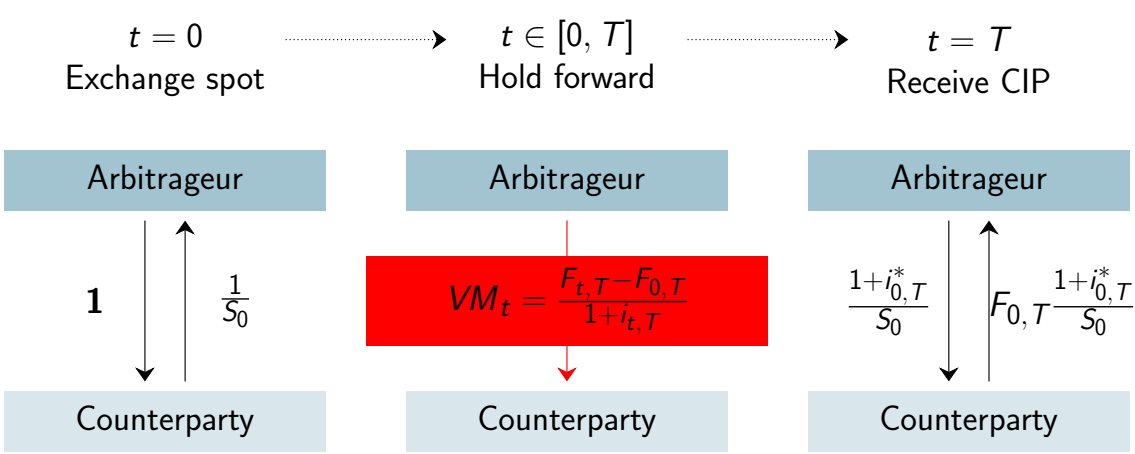


Figure: Assets and liabilities transferred into a FTP book (Tumasyan, 2012)

Variation margin payments require internal liquidity



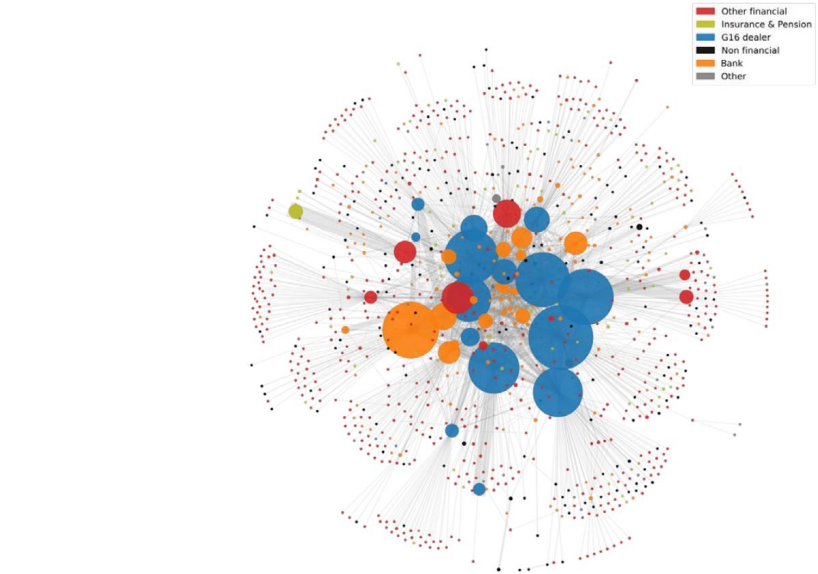


Figure: Counterparty network of gross notionals in EUR/USD FX forwards (Abad et al., 2016)

Main regression

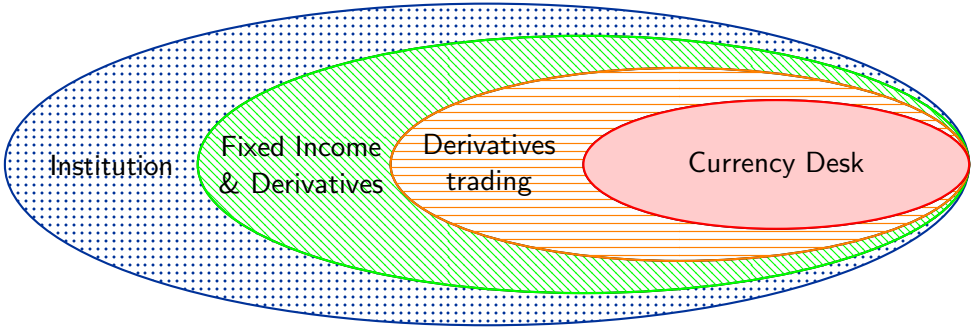


Figure: Partition of the balance-sheet of a major broker dealer into different levels.

$$\Delta b_{i,t} = \sum_{j \in \mathbb{J}} \beta_j \times \sum_{d(i) \in \mathbb{D}(i)} \omega_{d(i)} \Delta P\&L_{d(i),j,t-1} + controls_t + \epsilon_{i,t}$$

Main regression

$$\Delta b_{i,t} = \sum_{j \in \mathbb{J}} \beta_j \times \sum_{d(i) \in \mathbb{D}(i)} \omega_{d(i)} \Delta P\&L_{d(i),j,t-1} + controls_t + \epsilon_{i,t}$$

where:

- $b_{i,t}$ is the basis swap at time t in currency pair i,
- $\mathbb{J}$ is the partition of the dealers' balance sheets into the foreign exchange derivatives trading desk, all the other derivatives, all the other fixed income activities, and the rest of the institution,
- $\mathbb{D}(i)$ is the set of dealers in currency pair i,
- $\omega_{d(i)}$ is the weight of dealer $d(i)$ in currency pair i (degree / share of total notionals)
- $P\&L_{d(i),j,t}$ is the profit and loss statement of dealer $d(i)$ at level j at time t.

(UK-)EMIR data: The Regulation

(United Kingdom) **European Market Infrastructure Regulation** (EMIR):

- Global push for transparency in derivatives markets post GFC (G20 and FSB)
- European Regulation 648/2012 + UK onshoring legislation post Brexit
- Technical standards (FCA, ESMA, with IOSCO, CPMI, GLEIF)
- Accesses: Bank of England and FCA
- Coverage: UK entities as (at least one) counterparty

3 pillars: **Clearing, Reporting, Risk mitigation**

Data cleaning

Big data: ∼30 million open positions every day, with 129 fields by report for each counterparty.

(UK-)EMIR data: The Regulation

(United Kingdom) **European Market Infrastructure Regulation** (EMIR)

3 pillars:

- **Clearing obligation** for standardized contracts
- **Reporting obligation** of all the derivative contracts to a TR
- **Risk mitigation** techniques for non-centrally cleared contracts

Data cleaning

Big data: ∼30 million open positions every day, with 129 fields by report for each counterparty.

(UK-)EMIR data: The Regulation

(United Kingdom) **European Market Infrastructure Regulation** (EMIR)

3 pillars: **Clearing, Reporting, Risk mitigation**

Data cleaning

- Usually only based on quotes, or order-flow from one dealer for a short period of time
- Even with EMIR, usually only one TR, and always only based on one report
- Transaction-based (LEIs, P, Q), representative universe given London's role in FX trading

Big data: ∼30 million open positions every day, with 129 fields by report for each counterparty.

(UK-)EMIR data: The Regulation

(United Kingdom) **European Market Infrastructure Regulation** (EMIR)

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Big data: ∼30 million open positions every day, with 129 fields by report for each counterparty.

(UK-)EMIR data: Comparison to the BIS data

	(UK-)EMIR	BIS Triennial Survey	BIS OTC derivatives
Coverage	All derivatives	FX spot, FXD, IRD	OTC derivatives
Scope	All UK-based entities	"Reporting dealers" $>$ 50 jurisdictions	Dealer banks 12 jurisdictions
Stock or Flow	Stock and Flow	Flow	Stock
Level	Individual	Jurisdiction-sector	Jurisdiction-sector
Consolidation	None	None	Worldwide excluding intragroup
Frequency	Daily	Every 3 years	Every 6 months
Access	Confidential Some public aggregates	Public (jurisdiction-sector)	Public (sector)

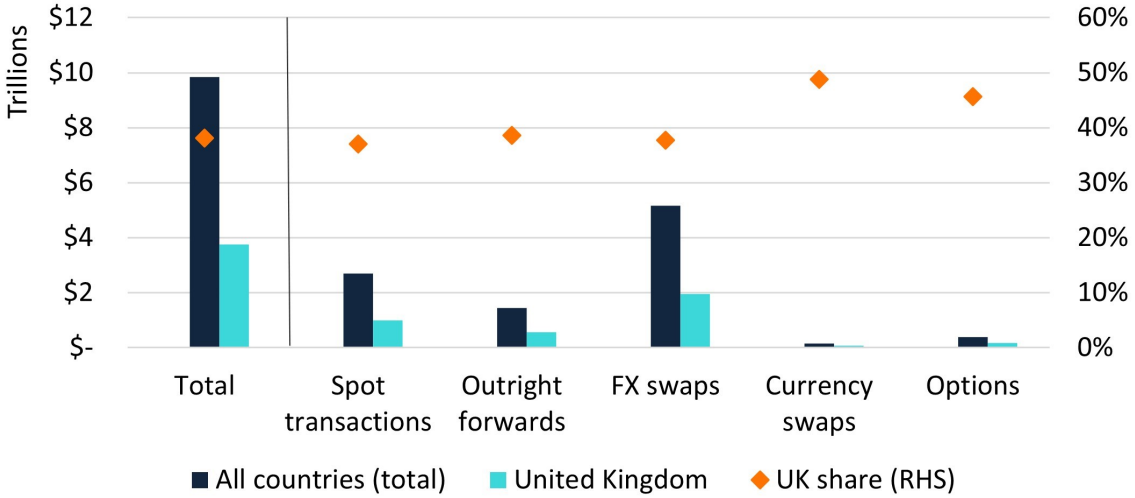


Figure: **Daily average turnover** (OTC foreign exchange "net-gross", BIS, 2022)

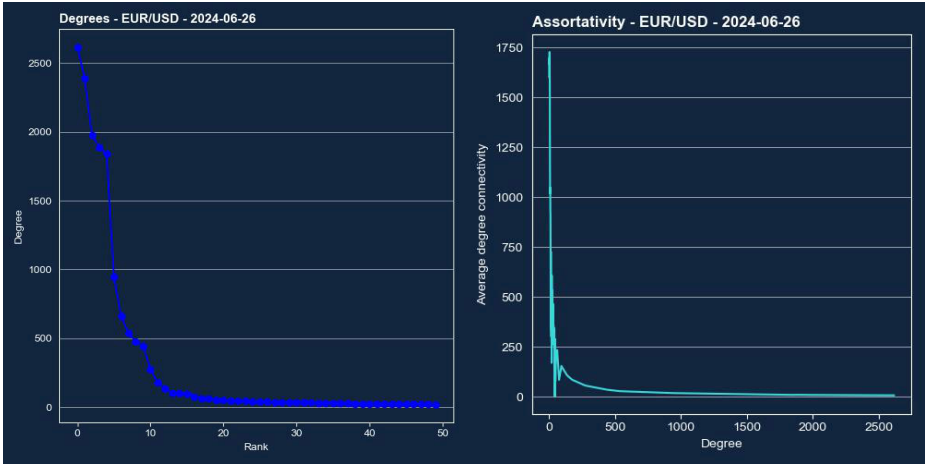
3 ways of using the data: (1) CIP deviations

$$b_T = \left[\left(1 + \frac{i_{0,T}}{100} \right) - \left(1 + \frac{i^*_{0,T}}{100} \right) \left(\frac{F_{0,T}}{S_0} \right)^{\frac{360}{T}} \right]$$

where:

- b_T is the contract level basis,
- $F_{0,T}$ is the contract-level forward rate,
- $i_{0,T}$ and $i^*_{0,T}$ zero coupon OIS curves from LSEG, interpolated with cubic splines,
- S_0 is the spot rate of the execution date,
- we use the convention Actual/360.

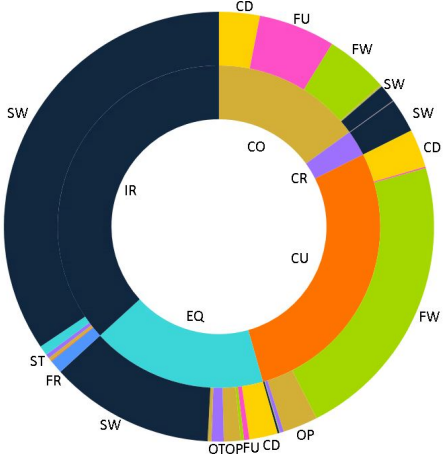
3 ways of using the data: (2) Identify the intermediaries



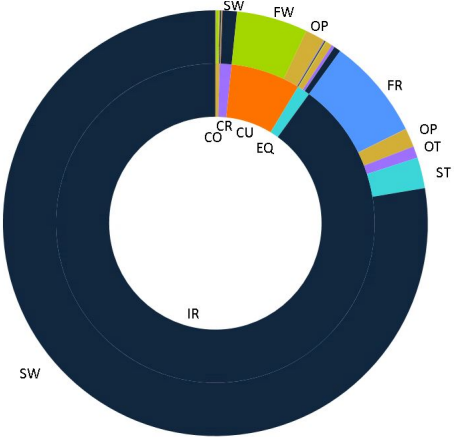
(a) Distribution of degrees (ranked)

(b) Assortativity

3 ways of using the data: (3) Compute the liquidity shocks



(a) Distribution of open positions (24 million)



(b) Distribution of notional values (USD782 trillion)

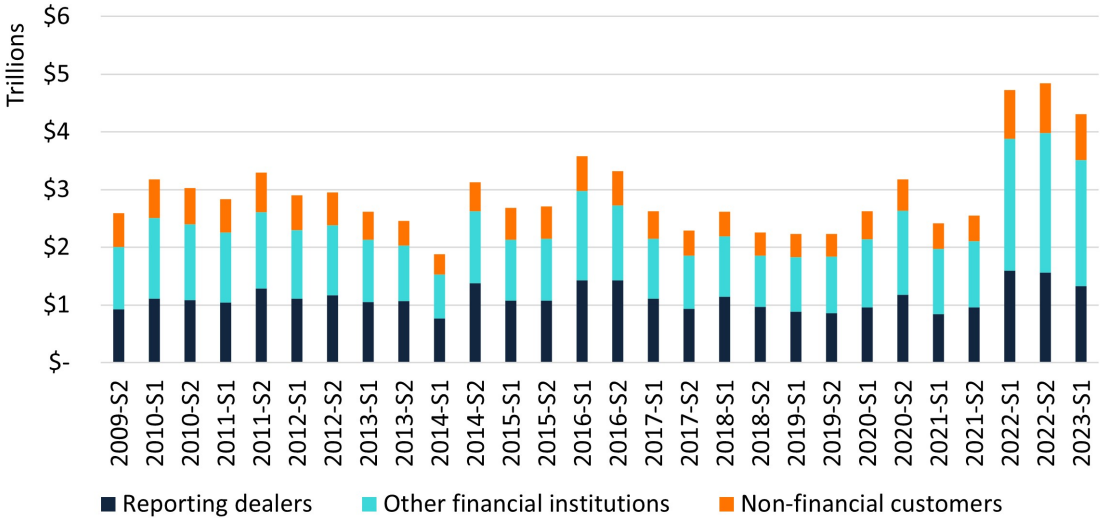


Figure: **FX derivatives gross market value** (forwards, FX swaps and currency swaps, BIS, S1-2023)

Policy implications

- If there are limits to arbitrage in the FX derivative market, then these limits might be significant in other, less liquid markets with a similar structure.
- The market microstructure can help reconcile macroeconomics and asset pricing, alternative to quantity-based approaches.
- The granularity of dominant intermediaries can be a source of systemic risk.

Conclusion

- Tentative to bridge the gap between the intermediary asset pricing theory and its empirical evaluations.
- Focus on internal liquidity frictions as a proxy for the SDF of intermediaries.
- Use the CIP deviations as a suitable laboratory to test our hypothesis.
- Document the market microstructure of FX forwards.

Feedback on the approach and suggestions are very welcome!

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Motivation - International Macroeconomics

Motivation: CIP violations are persistent and significant, matters for international macro.

A project in intermediary asset pricing... Why does it matter for (international) macro?

- MP setting: independence, transmission, imported inflation, export competitiveness
- LOOP violation in one market can propagate in other markets
- Important policy question (e.g. 2017 BIS Symposium: *CIP - RIP?*): If CIP violations reflect imbalances and financial frictions, might require CB swap lines
- CIP deviations might distort market decisions and the real economy, e.g. (bond) issuances

Key messages:

- The FX rate is one of the main variables in the economy
- The FX market is one of the most liquid in the world (trading continously 24h/6d). If there are limits to arbitrage there, then the limits might be significant in other, less liquid markets with a similar structure

Motivation - International Macroeconomics



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‘Reverse Yankee’ deals boom as Europe’s low borrowing costs lure US groups

American companies issue €30bn in euro-denominated bonds as investor demand for European corporate debt grows

(UK-)EMIR data: Reporting

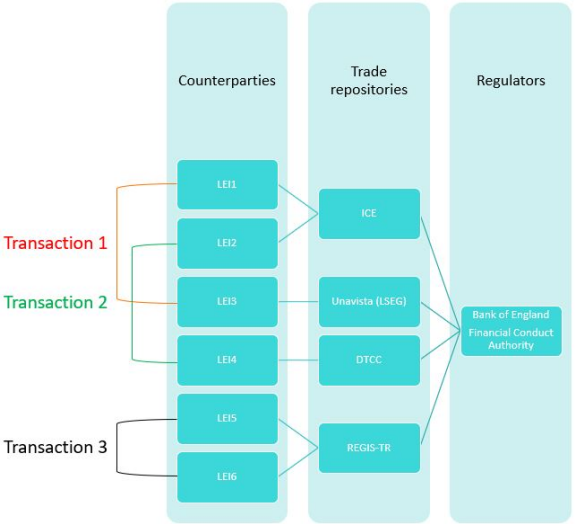


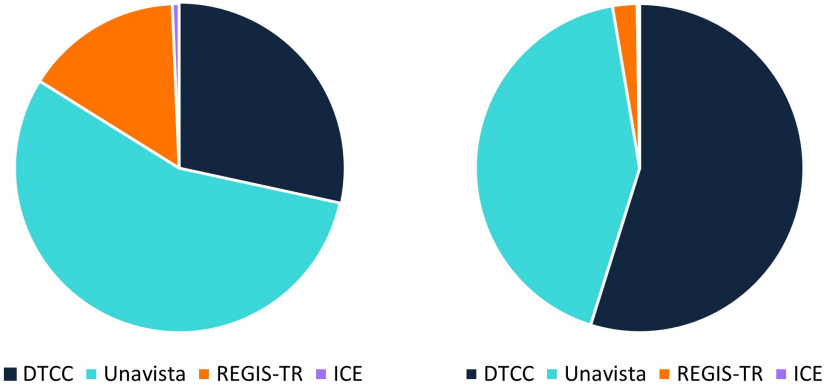
Illustration of the reporting process:

- 2 counterparties enter a derivative contract (suppose both UK-based)
- Each counterparty chooses a TR
- Regulators receive data from TRs

Double reporting:

+ Can compare the 2 sides
– Risk of double counting

(UK-)EMIR data: Trade repositories

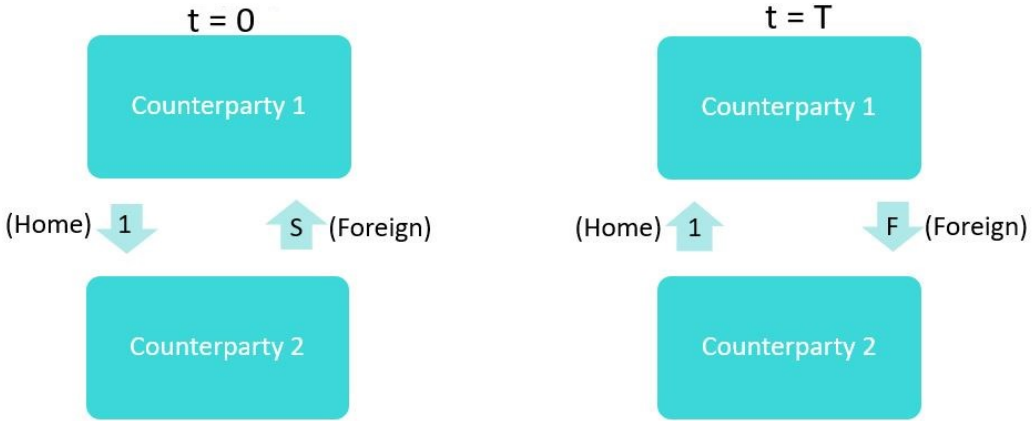


(a) Distribution of open positions

(b) Distribution of notional values

Figure: The market for reporting OTC FX derivatives

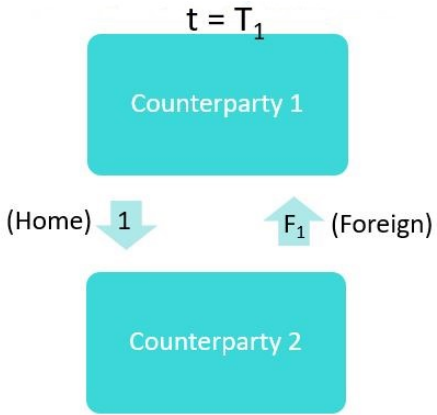
FX swaps as FX forwards? (1)

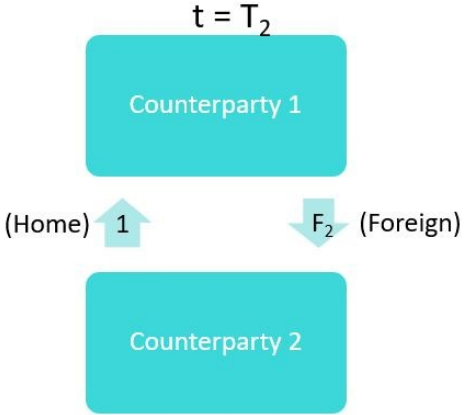


EMIR is about derivatives, not spot transactions.

- Only the forward leg is a derivative, hence the transaction reported as an FX forward.

FX swaps as FX forwards? (2)





This FX swap should be reported as a swap, given that it has two forward legs.

- Market convention is to report the two FX forwards independently.

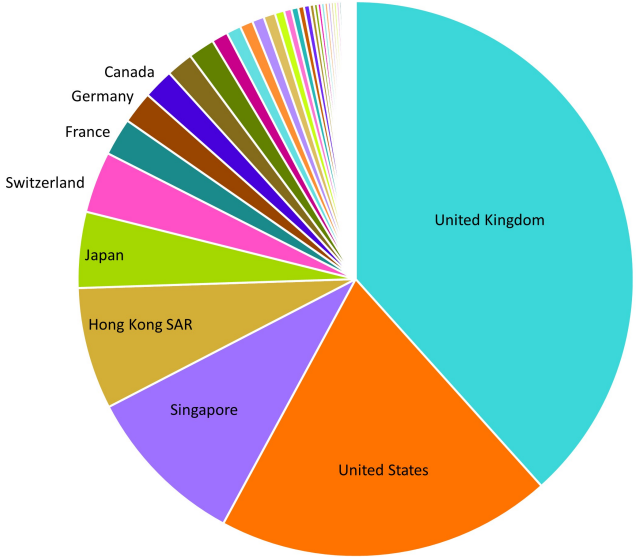


Figure: **Turnover by country** (Outright forwards, FX swaps and currency swaps, BIS, 2022)

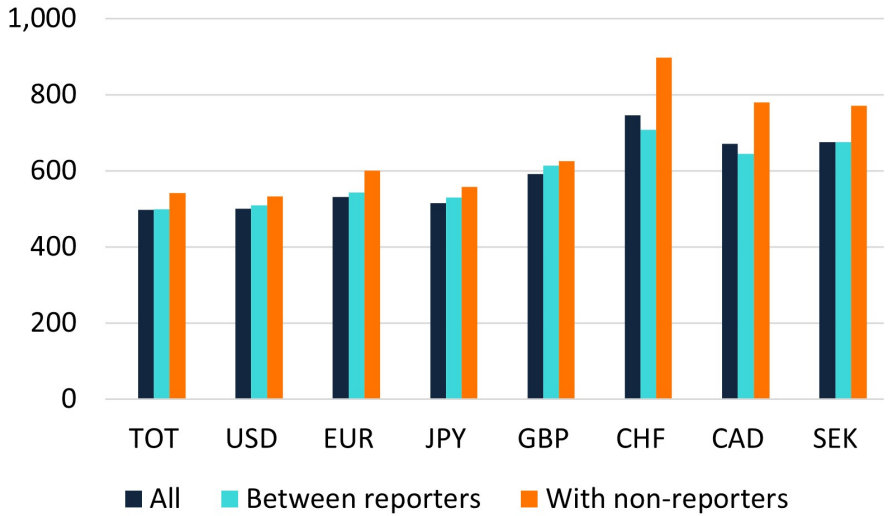


Figure: **HHI** (Outright forwards, forex swaps and currency swaps, BIS, S1-2023)

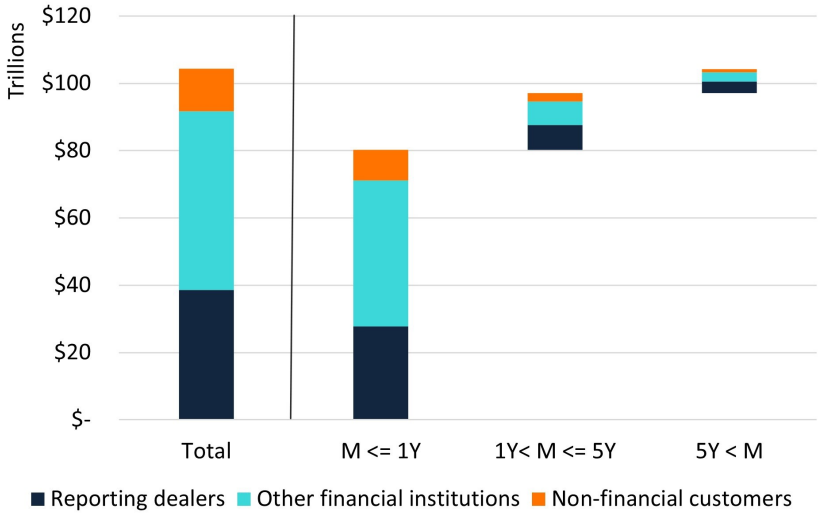


Figure: **Notionals outstanding** (Outright forwards, forex swaps and currency swaps, BIS, S1-2023)

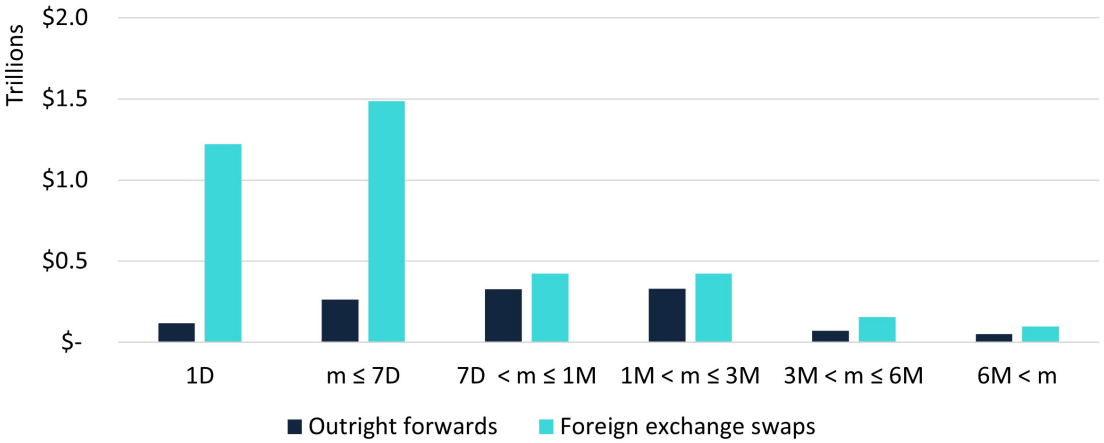


Figure: **Turnover by maturity** (Outright forwards, FX swaps and currency swaps, BIS, 2022)

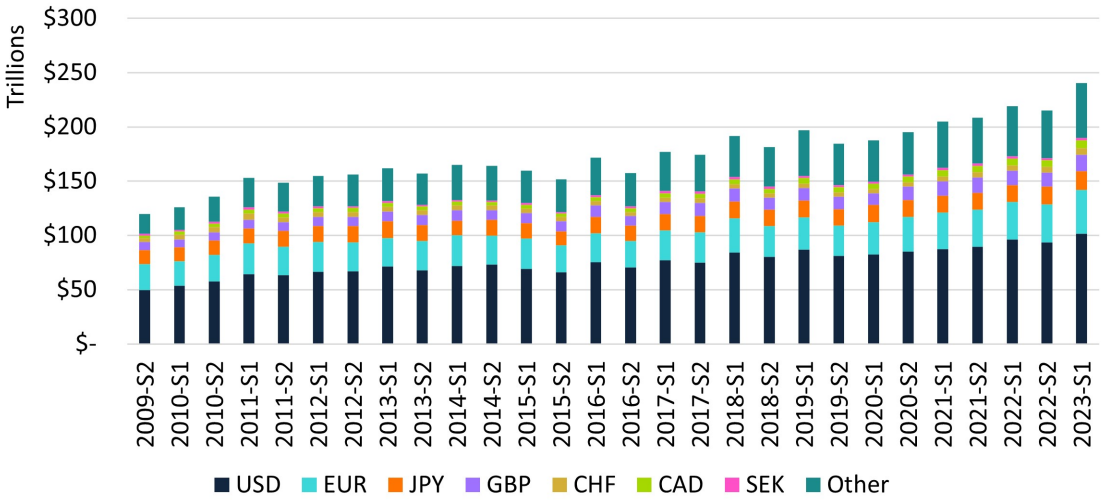


Figure: **Outstanding by currency** (Outright forwards, forex swaps and currency swaps, BIS, S1-2023)

Our contributions to the literature

We study internal liquidity and CIP violations, proxy for intermediaries' SDF

- Theory: Intermediary Asset Pricing / Limits to Arbitrage, FX determination
- Empirically, many limits to arbitrage can explain CIP violations
  - Adrian, Etula, and Muir (2014) vs He, Kelly, and Manela (2017)

➤ Because of limits to arbitrage, prices can move away from fundamentals

We empirically document the market structure for FX forwards

➤ The market is OTC and still intermediated by large dealers

We use individual P&L changes as shocks to explain aggregate prices

➤ Idiosyncratic shocks can have effects on macroeconomic variables

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We empirically document the market structure for FX forwards

- Theory: Search and matching for ad-hoc demand requires market dealers.
  - Alternatively, focus on the role of information.
- Empirics: Financial networks and price formation.
- Recent changes: move away from phone orders (!), RFQ / MTF / SI, algorithmic trading.
- The market is OTC and still intermediated by large dealers

We use individual P&L changes as shocks to explain aggregate prices

- Idiosyncratic shocks can have effects on macroeconomic variables

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- At the origins of slow moving capital: Internal capital constraints for trading desks with mark-to-market losses
- Arbitrage is driven by a limited number of market participants
  - Granular Instrumental Variables: Gabaix and Koijen (2020)
- ➤ Idiosyncratic shocks can have effects on macroeconomic variables